

packages/graph-core/src/change-classifier.ts

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Overview

`packages/graph-core/src/change-classifier.ts` adds a decision engine that classifies graph update types from a `ChangeAnalysis` (lines 1-2). It exports an `UpdateDecision` interface (lines 4-10) and a `classifyUpdate` function (lines 21-86) that chooses among `SKIP`, `PARTIAL_UPDATE`, `ARCHITECTURE_UPDATE`, and `FULL_UPDATE`.

Key changes

- **Imports:** `dirname` from `node:path` and `ChangeAnalysis` from `./fingerprint.js` (lines 1-2).
- **`UpdateDecision`** (lines 4-10): `action`, `filesToReanalyze`, `rerunArchitecture`, `rerunTour`, `reason`.
- **`classifyUpdate`** (lines 21-86):
 - `SKIP` when `structuralCount === 0`.
 - `FULL_UPDATE` if `structuralCount > 30` or `structuralCount / totalFilesInGraph > 0.5`.
 - `ARCHITECTURE_UPDATE` when `detectDirectoryChanges` is true or `structuralCount > 10`.
 - `PARTIAL_UPDATE` otherwise.
- **Helpers:**
 - `detectDirectoryChanges` (lines 94-113) checks for new/deleted top-level directories.
 - `topDirectory` (lines 119-124) returns the first path segment.
 - `summarizeChanges` (lines 129-143) builds a concise summary of new, deleted, and modified files.

Impact

- Centralizes update-decision logic; thresholds are explicit in the source.
- Decision logic is deterministic and documented in the comments.
- The module is isolated; consumers must import `UpdateDecision` and `classifyUpdate` to use it.

Risks & follow-ups

- `ChangeAnalysis` must correctly populate `structurallyChangedFiles`, `newFiles`, and `deletedFiles`; otherwise decisions may be wrong.
- `rerunTour` is set to `false` for `SKIP` and `true` for other actions; confirm downstream expectations.
- Edge cases (empty graph, 100% structural changes, directory-only changes) should be tested against the thresholds.

- `topDirectory` uses POSIX `/` separators; handling of Windows paths is unknown from the diff.